Reorganisation of an Outpatient Radiology Department

Simulation Modelling and Analysis (F000941) | 2025–2026

2. Statistical Analysis & Warm-up

2.1. Welch's Method and Warm-up Period

Before applying the batch means method, it is essential to determine and remove the initial warm-up period to eliminate transient initialization bias in the simulation output. To do this, we applied Welch's method across 10 distinct and representative experimental designs. These designs were chosen at random to ensure a diverse mix of urgent slots, scheduling rules, and timing strategies.

By plotting the moving average of the objective function for these designs, we visually inspected the point at which the output stabilized. When determining the universal warm-up length, we analyzed the worst-case scenario among these representative designs. Notably, configurations operating at near-full capacity (such as those with 18 urgent slots) were excluded from this specific baseline determination, as their extreme instability (which is further analyzed in later sections) skews the steady-state convergence.

Based on the remaining worst-case convergence, we observed a cutoff value of roughly 63 weeks. To maintain consistency and robustness across the majority of our experiments, we conservatively selected a universal warm-up period of **60 weeks**. This period is sufficient to remove initialization bias for approximately 90% of our tested scenarios, ensuring that only steady-state behaviour is considered in our final performance evaluations.

2.2. Batch Means Method

To conduct a statistically valid analysis of our steady-state simulation results, we applied the Batch Means Method. Because consecutive outputs in discrete-event simulations are autocorrelated, the batch means method groups observations into batches to form approximately independent data points.

Based on our worst-case scenario analysis, we determined that a batch size of $M = 80$ weeks ensures that the lag-based correlation falls below the 0.01 threshold. We sequentially determined that $L = 22$ batches are required to achieve the target precision. This yields a total simulation run length of:

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To establish a baseline of the natural system variance across
different configurations, we simulated 11 representative
experimental designs. The table below summarizes the 95% confidence
intervals, standard deviations, coefficient of variation (CV), and
relative precision ($\gamma$) for these designs using $M=80$ and
$L=22$. | Design | Avg_Obj ($\bar{D}$) | Var ($S^2$) | StdDev ($S$) |
Half-Width ($\epsilon$) | 95% CI | CV | Rel Precision ($\gamma$) | |
:--- | :--- | :--- | :--- | :--- | :--- | :--- | :--- | |
**S1-U14-R1** | 0.4762 | 0.0015 | 0.0386 | 0.0171 | [0.4591 ; 0.4933]
| 0.0811 | 0.0360 | | **S1-U18-R4** | 0.8340 | 0.0827 | 0.2876 |
0.1275 | [0.7065 ; 0.9615] | 0.3448 | 0.1529 | | **S3-U17-R2** |
0.5465 | 0.0117 | 0.1082 | 0.0480 | [0.4985 ; 0.5944] | 0.1981 |
0.0878 | | **S1-U11-R3** | 0.4542 | 0.0003 | 0.0182 | 0.0081 |
[0.4461 ; 0.4622] | 0.0401 | 0.0178 | | **S2-U17-R2** | 0.6162 |
0.0119 | 0.1091 | 0.0484 | [0.5678 ; 0.6646] | 0.1771 | 0.0785 | |
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S3-U14-R3 | 0.4468 | 0.0016 | 0.0398 | 0.0177 | [0.4291 ; 0.4645] | 0.0892 | 0.0395 | | **S3-U17-R3** | 0.5493 | 0.0118 | 0.1084 | 0.0481 | [0.5013 ; 0.5974] | 0.1974 | 0.0875 | | **S3-U13-R3** | 0.4320 | 0.0009 | 0.0307 | 0.0136 | [0.4184 ; 0.4457] | 0.0711 | 0.0315 | | **S1-U14-R4** | 0.4748 | 0.0015 | 0.0385 | 0.0171 | [0.4577 ; 0.4919] | 0.0812 | 0.0360 | | **S2-U16-R1** | 0.5657 | 0.0050 | 0.0704 | 0.0312 | [0.5345 ; 0.5969] | 0.1244 | 0.0552 | | **S2-U13-R4** | 0.5219 | 0.0009 | 0.0297 | 0.0132 | [0.5087 ; 0.5351] | 0.0570 | 0.0253 | The results indicate that while some configurations (e.g., S1-U11-R3) are highly stable with a relative precision of ~1.78%, others (e.g., S1-U18-R4) exhibit substantial natural variance with a relative precision exceeding 15%. This variance highlights the necessity of employing variance reduction techniques to ensure robust statistical comparisons between the designs. --- ## 3. Variance Reduction To obtain more accurate simulation estimates without increasing the computational burden, we implemented Variance Reduction using Antithetic Variables. For every standard simulation batch generated using random number sequence U, we generated an antithetic counterpart using $1-U$. These paired simulation outputs are negatively correlated. By averaging the outputs of the standard and antithetic runs ($Y_k = (X_k + X'_k)/2$), the overall variance is substantially reduced. ### Impact on Stable Systems For configurations that do not push the system to its capacity limits (e.g., 11 to 14 urgent slots), the antithetic pairing successfully reduced the variance by more than half without extending the simulation runtime. For example, in design **S1-U11-R3**, the variance dropped from 0.0003 to 0.0001, and the relative precision (γ) improved from 1.78% to 0.82%. Similarly, for **S1-U14-R1**, the variance dropped from 0.0015 to 0.0006 (γ improved from 3.6% to 2.18%). This mathematically confirms the efficacy of the antithetic variates approach for stabilizing estimates in normal operating conditions. ### The Phenomenon of Heavily Loaded Systems Conversely, the variance reduction technique exposed a critical vulnerability in heavily-loaded configurations (systems approaching 100% utilization, such as 17 or 18 urgent slots). For **S1-U18-R4**, the baseline mean was 0.8340. However, when the antithetic counterpart ($1 - U$) was run, the combined mean skyrocketed to 2.3745, and the variance heavily increased rather than decreasing. This happens because queue waiting times are convex relative to system utilization. When the antithetic sequence generates shorter inter-arrival times and longer service times, an already saturated queue completely breaks down, sending waiting times toward infinity. This finding is a powerful empirical proof that any design allocating 17 or 18 urgent slots is fundamentally unstable and highly susceptible to operational variability. Consequently, configurations with 18 or more urgent slots were excluded from the full experimental design. ### 3.1. Confidence Interval Estimation Since we are estimating the population mean based on a relatively small sample size ($n = L = 22$ batches), we used the

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$t$-distribution rather than the normal distribution. This
adjustment accounts for the additional uncertainty in estimating the
population standard deviation from a small sample. The general
formula for the confidence interval of the mean is given by:
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For the baseline design (**S1-U14-R1**) after applying antithetic variables, the 95% CI is `[0.4779 ; 0.4991]`, with `\bar{Y} = 0.4885`, `S_Y = 0.0240`, and `t_{0.025, 21} \approx 2.0796`. This narrow interval reflects the low variability across batch means after variance reduction, indicating a high degree of precision for stable configurations.

5. Comparative Analysis

5.1. Statistical Comparison of Top Designs

To determine whether differences between the top-performing designs are statistically significant, we examine whether their 95% confidence intervals overlap. Two designs can be considered significantly different if their confidence intervals do not overlap.

Comparing the top 4 designs (all S3-U11):

Design	95% CI
S3-U11-R4	[0.4271 ; 0.4352]
S3-U11-R2	[0.4280 ; 0.4361]
S3-U11-R1	[0.4284 ; 0.4365]
S3-U11-R3	[0.4304 ; 0.4385]

All four confidence intervals overlap substantially. We therefore cannot statistically distinguish between these four designs based on the current sample size. All S3-U11 configurations perform equivalently well, and the choice of appointment rule within this configuration is not statistically meaningful.

5.2. Comparison Between Strategies

The gap between Strategy 3 and the alternatives is statistically significant. Comparing the best S3 design (S3-U11-R4, CI: [0.4271; 0.4352]) against the best S1 design (S1-U11-R2, CI: [0.4491; 0.4566]): the upper bound of S3 (0.4352) is below the lower bound of S1 (0.4491), confirming that Strategy 3 is statistically superior to Strategy 1 for the same urgent slot count.

The superiority of Strategy 2 over Strategy 3 can be definitively rejected: the best S2 design (S2-U13-R4, CI: [0.5225; 0.5371]) lies entirely above the worst S3-U11 design (CI upper bound: 0.4385).

5.3. Recommended Configuration

Based on the full screening and comparative analysis, the recommended configuration is:

> ***Strategy 3 with 11 urgent slots (any appointment rule)***

The choice of appointment rule within this configuration does not significantly affect performance. If a single rule must be selected, **Rule 4 (Benchmarking)** achieves the lowest point estimate (OV = 0.4311), followed closely by Rule 2 (Bailey-Welch, OV = 0.4321).

This recommendation represents a change from the current schedule (Strategy 1, 14 urgent slots, Rule 1), which scores OV = 0.4880 — approximately **11% worse** than the best design.

5.4. Secondary Objectives

The assignment specifies that elective scan waiting time and overtime serve as secondary objectives for tiebreaking. Since the top-performing designs (S3-U11, all rules) are statistically indistinguishable on the primary objective, a formal tiebreak using secondary objectives could be applied. This analysis is left as a next step.